

Wei-Kai Li Academic CV

Professor / Doctoral Supervisor / Taishan Scholars Young Expert

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School of Computer Science and Artificial Intelligence, Shandong Jianzhu University / School of Mathematics and Statistics, Chongqing Jiaotong University / Panoramic Medical Imaging Center



Education

1. 2018–2022 / PhD / Computer Science and Technology / Nanjing University of Aeronautics and Astronautics
2. 2015–2018 / Master's / Computer Technology and Applications / Chongqing Jiaotong University
3. 2011–2015 / Bachelor's / Information and Computing Science / Chongqing Jiaotong University

Research Projects

1. Unsupervised Domain Adaptation beyond Domain Alignment and Its Extensions / NSFC Young Scientists Fund / Principal Investigator
2. Advanced School in Applied Machine Learning / NSFC International Cooperation and Exchange / Principal Investigator
3. Multimodal Neurophysiological Mechanisms of Heterogeneous Brain Networks in Depression and Digital-Twin Closed-Loop TMS Robotics / Shandong Major Basic Research Project / Principal Investigator
4. Exploration of Unsupervised Domain Adaptation beyond Domain Alignment / MIIT Key Laboratory Open Project / Principal Investigator
5. Early Identification of Parkinson's Disease Using PET/MR Multimodal Brain Networks / Hubei Key Laboratory of Molecular Imaging / Principal Investigator
6. Multiscale Analysis of Autism Using Multimodal Imaging / Chongqing Education Commission Young Research Fund / Principal Investigator
7. Brain Network Disease Diagnosis Based on Multicenter Transfer Learning / Chongqing Natural Science Foundation General Project / Principal Investigator
8. Probabilistic Synchronization and Secure Control of Stochastic Complex Networks / Chongqing Natural Science Foundation Joint Fund / Researcher
9. Investigating and Extending the Effectiveness of Self-Supervised Learning / NSFC General Program / Researcher

Publications

1. 2026 / Universum Inspired Graph Contrast Learning for Major Depressive Disorder Identification: In Functional Brain Network View. H Qin, G Hong, D Ma, Z Wang, L Zhao, W Li, X Xu. CNS Neuroscience & Therapeutics 32 (9), e71158, 2026
2. 2026 / Multi-Source Multi-View Graph Domain Adaptation with Hyperbolic Residual Encoding for Cross-Site MDD Identification from rs-fMRI. Z Zheng, X Chen, H Jiang, R Tian, Q Cai, J Liu, X Chen, W Li, Y Wang. arXiv preprint arXiv:2607.29531, 2026
3. 2026 / Domain generalization meets mutual bi-attention: A novel approach to drug-target prediction. J Yang, W Li, K Qian, Z Cheng. Neurocomputing, 134626, 2026
4. 2026 / Enhancing Fatigue Detection Through Heterogeneous Multisource Data Integration and Cross-Domain Modality Imputation. L Cui, Y Wu, Y Tang, W Li. IEEE Transactions on Computational Social Systems, 2026
5. 2026 / Hierarchical neighbor integration graph attention network for autism spectrum disorder diagnosis. D Ma, L Peng, L Zhang, W Li, X Gao. Frontiers in Psychiatry 17, 1846996, 2026
6. 2026 / TumorAL: Evidence-aware active learning for 3D tumor segmentation. H Wang, J Leng, S Li, Y Zhao, W Li, W Li, X Gao. Neurocomputing, 134239, 2026

7. 2026 / Identification of Major Depressive Disorder Using Multiple Functional Connection Patterns. Y Ruan, L Guan, L Peng, L Zhao, Y Huang, W Li, X Gao. *CNS Neuroscience & Therapeutics* 32 (6), e70951, 2026
8. 2026 / Electrocardiographic Alterations Combined with Hematological, Biochemical, and Metabolic Profiles Predict Prognosis in Kawasaki Disease. Q Wang, W Li, J Wan, L Wei, Y Xia, Y Hua, K Zhou, D Qie, W Li, Y Li. *Journal of Cardiovascular Development and Disease* 13 (6), 228, 2026
9. 2026 / Partial domain adaptation enables cross domain cell type annotation between scRNA-seq and snRNA-seq. X Chen, Q Zou, Q Cai, X Chen, W Li, Y Wang. *PLOS Computational Biology* 22 (5), e1014223, 2026
10. 2026 / Bridging the modality reliability gap in drug-target interaction prediction via a confidence-aware multimodal fusion framework. J Yang, J Zhang, K Qian, Q Yang, W Li, Z Cheng. *Proceedings of the AAAI Conference on Artificial Intelligence* 40 (32), 27529 ..., 2026
11. 2026 / Test-time few-shot object detection via dynamic prototype fusion. Y Wu, Y Li, H Wei, W Li, Y Tang. *IEEE Transactions on Cybernetics*, 2026
12. 2026 / Machine learning-based assessments of perioperative features in determining long-term conduct block post-transcatheter closure of ventricular septal defect. Q Wang, Y Xia, L Wei, P Yang, K Zhou, Y Hua, W Li, Y Li. *Cardiology in the Young*, 1-11, 2026
13. 2025 / Attention Transfer Based Hybrid Knowledge Distillation for Multimodal Brain Tumor Segmentation. H Ma, HS Gan, H Mi, P Xu, Y Liu, W Li. *2025 IEEE International Conference on Big Data (BigData)*, 7503-7510, 2025
14. 2025 / More insights into disruption and decoupling of individual metabolic connectomes in Parkinson's disease. D Li, J Yao, W Li, Z Guo, X Lv, L Chen, J Shi, YC Chen, J Ye. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, 111578, 2025
15. 2025 / Incorporating Negative Node Information in Graph Transformer for MDD Diagnosis. H Song, H Qin, W Li. *2025 International Conference on Cyber-Physical Social Intelligence (CPSI)*, 1-6, 2025
16. 2025 / MVSSL-GT: A Graph Transformer Method with Multi-View Contrastive Self-Supervised Learning. Y Lu, N Tian, W Li. *2025 International Conference on Cyber-Physical Social Intelligence (CPSI)*, 1-6, 2025
17. 2025 / Low-light image enhancement using dual cross attention. Y Ruan, H Ma, D Ma, W Li, X Wang. *Engineering Applications of Artificial Intelligence* 159, 111501, 2025
18. 2025 / Auto-weighted projective one-step multi-view clustering. X Mou, W Li, B Li, J Hu. *Neurocomputing*, 131960, 2025
19. 2025 / Revisiting Multi-Modal Alignment: In Distribution View. W Li, N Tian, Y Li, Y Tang. *2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC ...)*, 2025
20. 2025 / Quantitative susceptibility mapping shows alterations of brain iron content in children with autism spectrum disorder: a whole-brain analysis. X Xu, Y Li, H Lan, N Ding, W Li, G Zheng, X Song. *BMC psychiatry* 25 (1), 826, 2025
21. 2025 / Cross-modal medical image generation from MRI to PET using robust generative adversarial network. Y Yang, B Li, W Cao, X Chen, W Li. *Expert Systems with Applications*, 129287, 2025
22. 2025 / Leveraging multi-source and heterogeneous signals for fatigue detection. L Cui, Y Wu, T Ying, W Li. *arXiv e-prints*, arXiv: 2507.16859, 2025
23. 2025 / PSMA PET/CT based multimodal deep learning model for accurate prediction of pelvic lymph-node metastases in prostate cancer patients identified as candidates for extended Q Ma, B Chen, R Seifert, R Zhou, L Xiao, J Yang, A Rominger, K Shi, W Li, *European Journal of Nuclear Medicine and Molecular Imaging* 52 (6), 2063-2075, 2025
24. 2025 / Timestamp calibration for time-series single cell rna-seq expression data. X Chen, S Lin, X Chen, W Li, Y Li. *Journal of Molecular Biology* 437 (9), 169021, 2025
25. 2025 / Iterative neural networks for improving memory capacity. X Chen, D Lin, Z Li, W Li. *Neural Networks* 182, 106936, 2025
26. 2024 / The hidden Markov model reveals the changes in brain dynamics among patients with end-stage renal disease under different dialysis methods. D Sun, W Li, H Yu, X Chen, C Zhang. *Journal of Cyber-Physical-Social Intelligence* 3 (1), 10-10, 2024
27. 2024 / An Improved Algorithm for Spiking Neural Networks with Multi-Scale Attention Coding. S Chen, X Chen, W Li. *2024 International Conference on Cyber-Physical Social Intelligence (ICCSI)*, 1-6, 2024

28. 2024 / Improved domain-adversarial neural networks with mixup and pseudo-labeling. X Xiao, X Chen, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
29. 2024 / Dynamic Graph Contrastive Learning for Neuro Disease Identification using Resting-State fMRI. Y Jiang, W Li, X Chen. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
30. 2024 / Synthesizing A β -PET based on multi-modal neuroimaging fusion for pathological diagnosis of Alzheimer's disease. Y Yang, B Li, W Li, H Chen, W Cao. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
31. 2024 / Universum Inspired Graph Contrast Learning for Autism Spectrum Disorder Identification. H Qin, X Xu, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
32. 2024 / Enhancing Functional Brain Network Construction for Neurological Disorder Diagnosis through Modularity-Driven Regularization. D Sun, X Chen, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
33. 2024 / Multi-view comparative self-supervised graph learning for neuro-disease diagnosis using functional brain networks. N Tian, X Xu, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
34. 2024 / Topology-Enhanced Multi-Modal Residual Graph Convolutional Network for Autism Spectrum Disorder Identification. J Deng, L Guan, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
35. 2024 / Efficient and accurate capsule networks with b-spline-based activation functions. L Mou, X Xiao, W Cao, W Li, X Chen. 2024 International Conference on New Trends in Computational Intelligence ..., 2024
36. 2024 / The characteristic patterns of individual brain susceptibility networks underlie Alzheimer's disease and white matter hyperintensity-related cognitive impairment. H Chen, J Xu, W Li, Z Hu, Z Ke, R Qin, Y Xu. *Translational Psychiatry* 14 (1), 177, 2024
37. 2024 / TIDE: Test-time few-shot object detection. W Li, H Wei, Y Wu, J Yang, Y Ruan, Y Li, Y Tang. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 54 (11), 6500-6509, 2024
38. 2024 / Multipattern graph convolutional network-based autism spectrum disorder identification. W Zhou, M Sun, X Xu, Y Ruan, C Sun, W Li, X Gao. *Cerebral Cortex* 34 (3), bhae064, 2024
39. 2024 / Limbic/paralimbic connection weakening in preschool autism-spectrum disorder based on diffusion basis spectrum imaging. T Yi, W Li, W Wei, G Wu, G Jiang, X Gao, K Jin. *The European journal of neuroscience*, 2024
40. 2024 / Robust support vector machine based on sample screening. J Guo, W Li, J Hu. *The 14th International Conference on Information Science and Technology*, 2024
41. 2024 / Altered intra-and inter-network connectivity in autism spectrum disorder. R Zhou, C Sun, M Sun, Y Ruan, W Li, X Gao. *Aging* 16, 2024
42. 2024 / Machine learning models using symptoms and clinical variables to predict coronary artery disease on coronary angiography. Y Yu, W Li, J Wu, X Hua, B Jin, H Shi, Q Chen, J Pan. *Advances in Interventional Cardiology/Postępy w Kardiologii Interwencyjnej* ..., 2024
43. 2024 / Towards an accurate autism spectrum disorder diagnosis: multiple connectome views from fMRI data. J Yang, X Xu, M Sun, Y Ruan, C Sun, W Li, X Gao. *Cerebral Cortex* 34 (1), bhad477, 2024
44. 2024 / Effects of individual metabolic brain network changes co-affected by T2DM and aging on the probabilities of T2DM: protective and risk factors. YL Li, JJ Wu, WK Li, X Gao, D Wei, X Xue, XY Hua, MX Zheng, JG Xu. *Cerebral Cortex* 34 (1), bhad439, 2024
45. 2024 / Topological properties analysis and identification of mild cognitive impairment based on individual morphological brain network connectome. X Xu, P Chen, W Li, Y Xiang, Z Xie, Q Yu, Y Tang, P Wang. *Cerebral Cortex* 34 (1), bhad450, 2024
46. 2023 / Wavelet transform-based frequency self-adaptive model for functional brain network. Y Ding, X Xu, L Peng, L Zhang, W Li, W Cao, X Gao. *Cerebral Cortex* 33 (22), 11181-11194, 2023
47. 2023 / PageRank algorithm based on dynamic damping factor. Z HaoLin, H Jin, L WeiKai. 2023 International Conference on Cyber-Physical Social Intelligence (ICCSI ..., 2023
48. 2023 / An improved AdaBoost method in imbalanced data Learning. T Li, X Chen, W Li. 2023 International Conference on Cyber-Physical Social Intelligence (ICCSI ..., 2023
49. 2023 / Minimum error entropy high-order extended Kalman filter. Z Cheng, X Chen, H Li, W Li, D Lin. 2023 International Conference on Cyber-Physical Social Intelligence (ICCSI ..., 2023

50. 2023 / New insights into the disorder of brain connectivity in schizophrenia. W Li, MM Al-Amin, A Nair. Frontiers in Neuroimaging 2, 1266695, 2023
51. 2023 / The reconfiguration pattern of individual brain metabolic connectome for Parkinson's disease identification. W Li, Y Tang, L Peng, Z Wang, S Hu, X Gao. MedComm 4 (4), e305, 2023
52. 2023 / TIToK: A solution for bi-imbalanced unsupervised domain adaptation. Y Wang, Q Chen, Y Liu, W Li, S Chen. Neural Networks 164, 81-90, 2023
53. 2023 / Clinical application of machine learning methods in psychiatric disorders. X Liu, Z Xu, W Li, Z Zhou, X Jia. Frontiers in psychiatry 14, 1209615, 2023
54. 2023 / Functional and structural brain network construction, representation and application. W Li, Z Wang, S Hu, C Chen, M Liu. Frontiers in neuroscience 17, 2023
55. 2022 / Unsupervised domain adaptation with progressive adaptation of subspaces. W Li, S Chen. Pattern Recognition 132, 108918, 2022
56. 2022 / Jacobian norm for unsupervised source-free domain adaptation. W Li, M Cao, S Chen. arXiv preprint arXiv:2204.03467, 2022
57. 2022 / Machine learning-derived multimodal neuroimaging of presurgical target area to predict individual's seizure outcomes after epilepsy surgery. Y Tang, W Li, L Tao, J Li, T Long, Y Li, D Chen, S Hu. Frontiers in Cell and Developmental Biology 9, 669795, 2022
58. 2022 / Human-Guided Functional Connectivity Network Estimation for Chronic Tinnitus Identification: A Modularity View. WK Li, YC Chen, XW Xu, X Wang, X Gao. IEEE journal of biomedical and health informatics, 2022
59. 2022 / Partial Domain Adaptation without Domain Alignment. W Li, S Chen. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2022
60. 2022 / Machine learning based on the multimodal connectome can predict the preclinical stage of Alzheimer's disease: a preliminary study. H Chen, W Li, X Sheng, Q Ye, H Zhao, Y Xu, F Bai, European Radiology 32 (1), 448-459, 2022
61. 2021 / Multiple connection pattern combination from single-mode data for mild cognitive impairment identification. W Li, X Xu, Z Wang, L Peng, P Wang, X Gao. Frontiers in Cell and Developmental Biology 9, 782727, 2021
62. 2021 / Morphological, structural, and functional networks highlight the role of the cortical-subcortical circuit in individuals with subjective cognitive decline. X Xu, T Wang, W Li, H Li, B Xu, M Zhang, L Yue, P Wang, S Xiao. Frontiers in Aging Neuroscience 13, 688113, 2021
63. 2021 / Characteristics of multimodal brain connectomics in patients with schizophrenia and the unaffected first-degree relatives. X Lin, WK Li, G Dong, Q Wang, H Sun, J Shi, Y Fan, P Li, L Lu. Frontiers in cell and developmental biology 9, 631864, 2021
64. 2020 / Leave zero out: Towards a no-cross-validation approach for model selection. W Li, C Geng, S Chen. arXiv preprint arXiv:2012.13309, 2020
65. 2020 / Effective and accurate diagnosis of subjective cognitive decline based on functional connection and graph theory view. X Xu, W Li, M Tao, Z Xie, X Gao, L Yue, P Wang. Frontiers in neuroscience 14, 577887, 2020
66. 2020 / Functional connectivity network estimation with an inter-similarity prior for mild cognitive impairment classification. W Li, X Xu, W Jiang, P Wang, X Gao. Aging (Albany NY) 12 (17), 17328, 2020
67. 2020 / The effect of jet lag on the human brain: A neuroimaging study. F Zhang, W Li, H Li, S Gao, JA Sweeney, Z Jia, Q Gong. Human brain mapping 41 (9), 2281-2291, 2020
68. 2020 / Group similarity constraint functional brain network estimation for mild cognitive impairment classification. X Gao, X Xu, X Hua, P Wang, W Li, R Li. Frontiers in Neuroscience 14, 165, 2020
69. 2020 / Feature selection and combination of information in the functional brain connectome for discrimination of mild cognitive impairment and analyses of altered brain patterns. X Xu, W Li, J Mei, M Tao, X Wang, Q Zhao, X Liang, W Wu, D Ding, Frontiers in aging neuroscience 12, 28, 2020
70. 2020 / ■■■■■■■■■■■■■■■■■■■■■■. ■■■■ ■■■■ ■■■■ ■■■■ ■■■■. ■■■■■■■■■■ 2 (1), 80-87, 2020
71. 2019 / Toward a better estimation of functional brain network for mild cognitive impairment identification: a transfer learning view. W Li, L Zhang, L Qiao, D Shen. IEEE journal of biomedical and health informatics 24 (4), 1160-1168, 2019

Personal Recognition

1. 2026 / CPSI Best Paper Award
2. 2025 / Taishan Scholars Young Expert
3. 2025 / CPSI Best Paper Award
4. 2025 / Outstanding Faculty Advisor, China Robot and AI Competition
5. 2023 / ICCSI Best Paper Finalist Award
6. 2022 / National Scholarship for Doctoral Students

Student Awards

1. 2026 / Jingxuan Cui and Coauthors: CPSI 2026 Best Paper Award
2. 2025 / First Prize, China Robot and AI Competition National Finals
3. 2025 / First Prize, China Robot and AI Competition Chongqing Selection
4. 2025 / National Third Prize, National Digital Art Design Competition
5. 2025 / MCM/ICM Honorable Mention and Meritorious Winner
6. 2025 / First Prize, China International College Students' Innovation Competition: Paying Xianzhi
7. 2025 / First Prize, China International College Students' Innovation Competition: Yingluo Zhizhen
8. 2025 / Second Prize, iCAN Campus Competition
9. 2025 / Third Prize, China Graduate Electronics Design Competition, Southwest Region
10. 2024 / MCM/ICM Honorable Mention and Meritorious Winner
11. 2024 / First and Third Prizes, China International College Students' Innovation Competition, Campus Round
12. 2024 / Second Prize, China Robot and AI Competition National Finals
13. 2024 / Second Prize, Chongqing Robot Competition
14. 2024 / First Prize, National Business Elite Challenge Finals
15. 2024 / Third Prize, National Intelligent Transportation Innovation and Entrepreneurship Competition
16. 2024 / Third Prize, China Graduate Electronics Design Competition
17. 2024 / Second Prize, University Entrepreneurship Competition
18. 2024 / First Prize, RAICOM Robot Developer Competition
19. 2023 / First and Second Prizes, National Undergraduate Mathematical Contest in Modeling, Chongqing
20. 2023 / Third Prize, China Internet+ College Students' Innovation and Entrepreneurship Competition, Campus Round
21. 2023 / Third Prize, China Software Cup Western Division